

VIMALRAJ MURUGAN

FRONTEND ENGINEER

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SUMMARY

Frontend Engineer with 4 years owning the full design to production using React, Next.js, and TypeScript applications. I like building motion-heavy, interactive experiences, reusable components and 3D web experiences like GSAP ScrollTrigger, Three.js, React Three Fiber and also blender, I care about the details: loading states, accessibility, performance. I've taken projects solo from design handoff to production, including a funded startup's investor-facing site and built an e-commerce platform

EDUCATION

Alpha College Arts and Science – Chennai, India

Aug 2017 - April 2020

Bachelor of Science (Computer Science), First Class

- Leadership: NCC Group Leader — led 25 cadets through drills, events, and field exercises

PROFESSIONAL EXPERIENCE

Freelance Frontend Engineer

Nov 2025 - Present

- Partnered with fast-paced startup teams to ship 4+ production web applications and an internal SaaS product end-to-end in React, Next.js, and TypeScript, interfacing directly with client technical teams from design handoff through production deploy

Manvasam (Startup) - B2B Coconut Export (Singapore) - Apr 2026 – Jun 2026 Freelance

- Delivered premium Design and developed B2B coconut processing platform for a Singapore based client, handling both UI/UX design and full frontend development
- Accelerated development by 30% using Claude Code to build Next.js components and turn Figma designs into reusable, type-safe code.
- Delivered fully responsive design with SEO optimization for an international audience (<https://manvasam.vercel.app>)

Modulus Housing - Freelance

Nov 2025 – Feb 2026

- Developed Next Js website for Modulus Housing, An IIT Madras startup that raised \$7.83M (₹70Cr) Series A from Kalaari Capital & Hero
- Developed interactive India map with hover-state data visualization showing buildings built per state
- Created scroll triggered crane animation ("Build in 30 Days") using After Effects, Bodymovin, Lottie
- Built animated brand sections with Framer Motion — delivering a high-conversion experience for investors and B2B stakeholders
- Built a reusable component library of 20+ typed React components covering layout, forms and Set up CI/CD with GitHub Actions and Vercel automating preview deploys on every pull request. (www.modulushousing.com)

New Wings Solutions – B2B Cinema Equipment Platform - Freelance

- Designed the interface in Figma before writing code, then built it in React and Next.js. Made the calls on layout, type, color, and spacing, and designed the states people skip: loading, empty, and error, not just the main flow.
- Set the visual direction: a dark cinema theme with editorial typography, and a set of SVG illustrations I drew for the product categories.
- Optimized live product search with useTransition and useDeferredValue so filtering a long list doesn't block the main thread or shift the layout while you type.
- Connected the contact and quote forms to Next.js API routes backed by Supabase, with validation on both the client and the server

CORE SKILLS

Languages

- TypeScript
- JavaScript (ES6+)
- HTML5
- CSS3

Frameworks

- React
- Next.js
- React Native
- Node.js

Motion Design & UI

- Tailwind CSS
- Framer Motion
- NativeWind
- Three.js / R3F
- After Effects + Lottie
- Responsive Design
- Figma (UI/UX)
- Blender
- GSAP, ScrollTrigger

Static & Real-time

- Zustand
- Rest APIs
- Redux

Tooling & CMS

- Contentful
- GitHub Actions
- Vercel
- Claude Code

- Built Tabs and Dialog as reusable components on top of React Context, handling the accessibility roles and ARIA labels in one place so every instance inherits them, and covered the components with Jest and React Testing Library.
- Added a Cmd+K command palette with fuzzy search and full keyboard navigation.
- Built dark and light themes that apply the saved preference before first paint, plus a language switcher for English, Tamil, and Hindi.
- Chose the rendering method per route (static, ISR, or server-side) based on how often the data changes, and kept initial load light with next/image and IntersectionObserver lazy loading.
- Live Link: <https://www.newwingsolutions.com>

BEYOND CODE

- Multi-disciplinary delivery across web, animation, and video including After Effects/Premiere Pro post-production (subtitles, edits, final cut) for a corporate client

Farm To Home - Singapore - (Mini E-commerce Platform) - Freelance

June 2026 - Present

- Architected and built a full-stack e-commerce platform from scratch using Next.js (App Router), TypeScript, Tailwind CSS, and Supabase (PostgreSQL, Auth, Storage), currently serving a live business with a 939-product catalog
- Designed a custom headless CMS enabling a non-technical business owner to independently manage all site content, products, and homepage sections with zero code involvement
- Integrated Stripe for payment processing with signature-verified webhooks, idempotency keys, and atomic race-condition-safe order state handling; built a secondary manual payment flow (PayNow) for regional preference
- Implemented a dual-authentication architecture with fully separated admin and customer sessions.
- Built a secure cart/checkout system with server-side price and quantity re-validation (Zod), dynamic coupon/tax/shipping calculation.
- Owned full infrastructure: DNS/domain configuration, CI/CD via GitHub + Vercel, transactional email delivery (Resend, custom SMTP, DKIM/SPF/DMARC), and database hosting
- Directed an AI-assisted development workflow (Claude Code) for rapid iteration, while independently reviewing all security-critical and production code
- Live Link: <https://farmtohome.sg>

New Wings Solutions – Chennai, India - Full Time (Startup) Oct 2022 - Sept 2025

Web Developer

- Shipped 8+ client websites and deployed using React, Next.js, and TypeScript and React Native mobile app with Clerk authentication in under 3 weeks, packaging a production-ready APK for on-site distribution at a school expo
- Translated complex Figma prototypes into pixel-perfect Next.js layouts, optimizing Core Web Vitals, SSR and image optimization in Next.js and Built type-safe REST APIs using TypeScript (Next.js, Node.js), achieving 50-75% faster data fetch speeds.
- Modeled custom 3D objects in Blender and brought them into a Next.js scene using Three.js / React Three Fiber.
- Built scroll-driven storytelling with GSAP ScrollTrigger the 3D model reshapes and re-angles as the user scrolls, turning scroll position into an animation timeline.
- Set up Contentful as a headless CMS so the client's team could edit content without touching code.

PERSONAL LEARNING PROJECT

AI Frontend Engineering

Sphere – Real-time Voice AI Interface

Next.js · TypeScript · WebSockets · Web Audio API · React Three Fiber · Node

- Explored WebSockets and Web Audio API to build a real-time voice interface
- Experimented with an audio-reactive 3D scene in React Three Fiber driven by live amplitude data